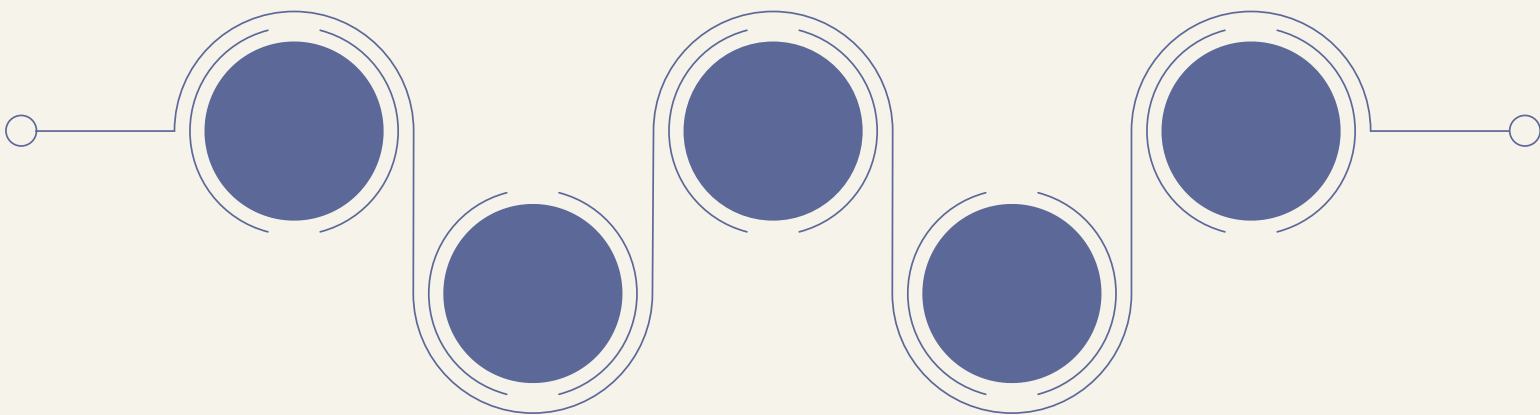




EARLY DROPOUT DETECTION

Eric Wang, Sam Chernoff, Joy Cheng, Grayson Stone

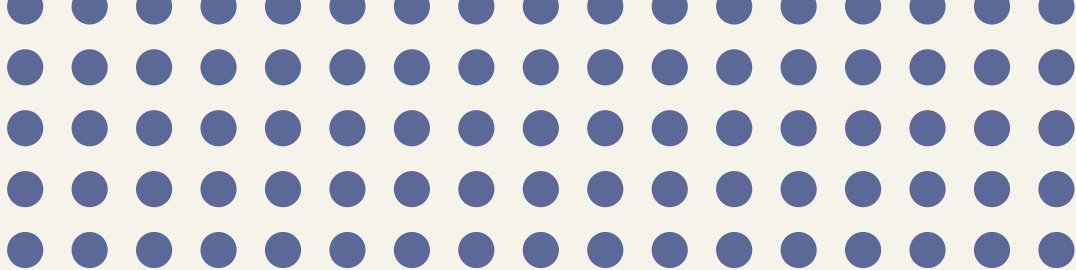




Overview

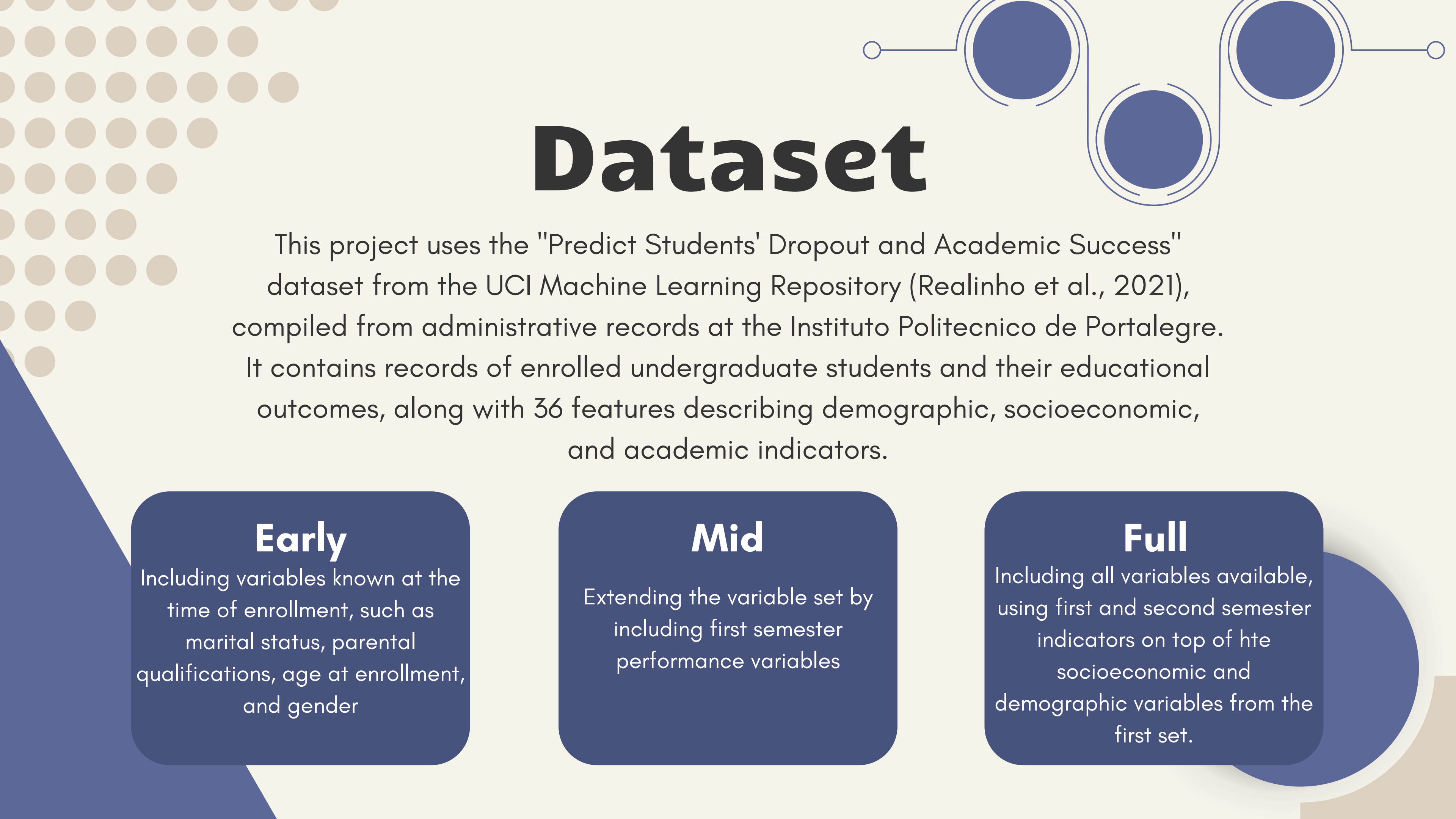
- **Background**
 - **Dataset**
 - **Methodology**
 - **Results**
 - **Conclusion**
 - **References**
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Background

- Dropping out of school has social costs:
 - Decreased labor market success (Education pays, 2024)
 - Criminal activity (Lochner, 2020)
 - Public finance burden (Lochner and Moretti, 2004)
 - Investment in education provides social benefit
 - The ability to predict when a student might drop out is crucial for intervention and prevention
 - We sought out to:
 - **Model predicting dropout**
 - Identify **key predictors**
 - Identify **when** data should be collected
 - Our model can be valuable in guiding data collection and education policy in the future
- 



Dataset

This project uses the "Predict Students' Dropout and Academic Success" dataset from the UCI Machine Learning Repository (Realinho et al., 2021), compiled from administrative records at the Instituto Politecnico de Portalegre. It contains records of enrolled undergraduate students and their educational outcomes, along with 36 features describing demographic, socioeconomic, and academic indicators.

Early

Including variables known at the time of enrollment, such as marital status, parental qualifications, age at enrollment, and gender

Mid

Extending the variable set by including first semester performance variables

Full

Including all variables available, using first and second semester indicators on top of the socioeconomic and demographic variables from the first set.

Methodology

Logistic Regression

- Linear baseline model
- Ridge/L2 shrinks coefficients toward zero
- Lasso/L1 can set coefficients to zero, enabling feature selection
- Trained on standardized features

Random Forest

- Ensemble of bootstrapped decision trees
- Captures nonlinear relationships and feature interactions
- Robust to outliers and class imbalance
- Used 100 trees, max depth 15, balanced class weights

XGBoost

- Gradient-boosted tree ensemble
- Sequentially improves by correcting prior errors
- Strong for structured data with complex interactions
- Used 100 estimators, depth 7, learning rate 0.1

Data split: 80% training, 20% test using stratified sampling.

Cross validation: Training set validated with 5-fold stratified CV.

Optimization metric: weighted F1, appropriate for imbalanced classes.

Final test metrics: weighted F1, accuracy, and ROC-AUC.



Results

Average Algorithm Performance

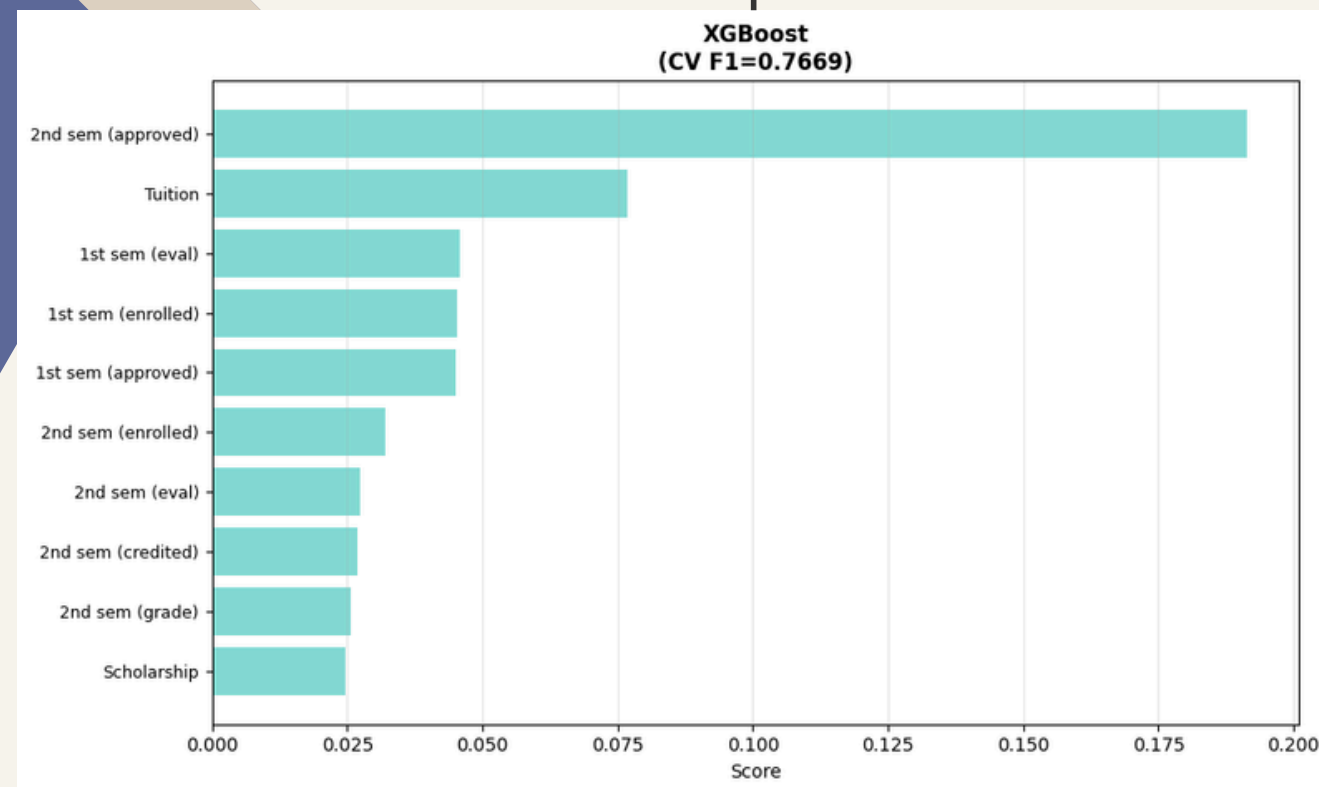
Algorithm	Test Accuracy	Test F1	Test ROC-AUC
LogReg (Lasso)	0.707	0.669	0.814
LogReg (Ridge)	0.71	0.678	0.82
Random Forest	0.688	0.69	0.832
XGBoost	0.711	0.699	0.838

Average Dataset Performance

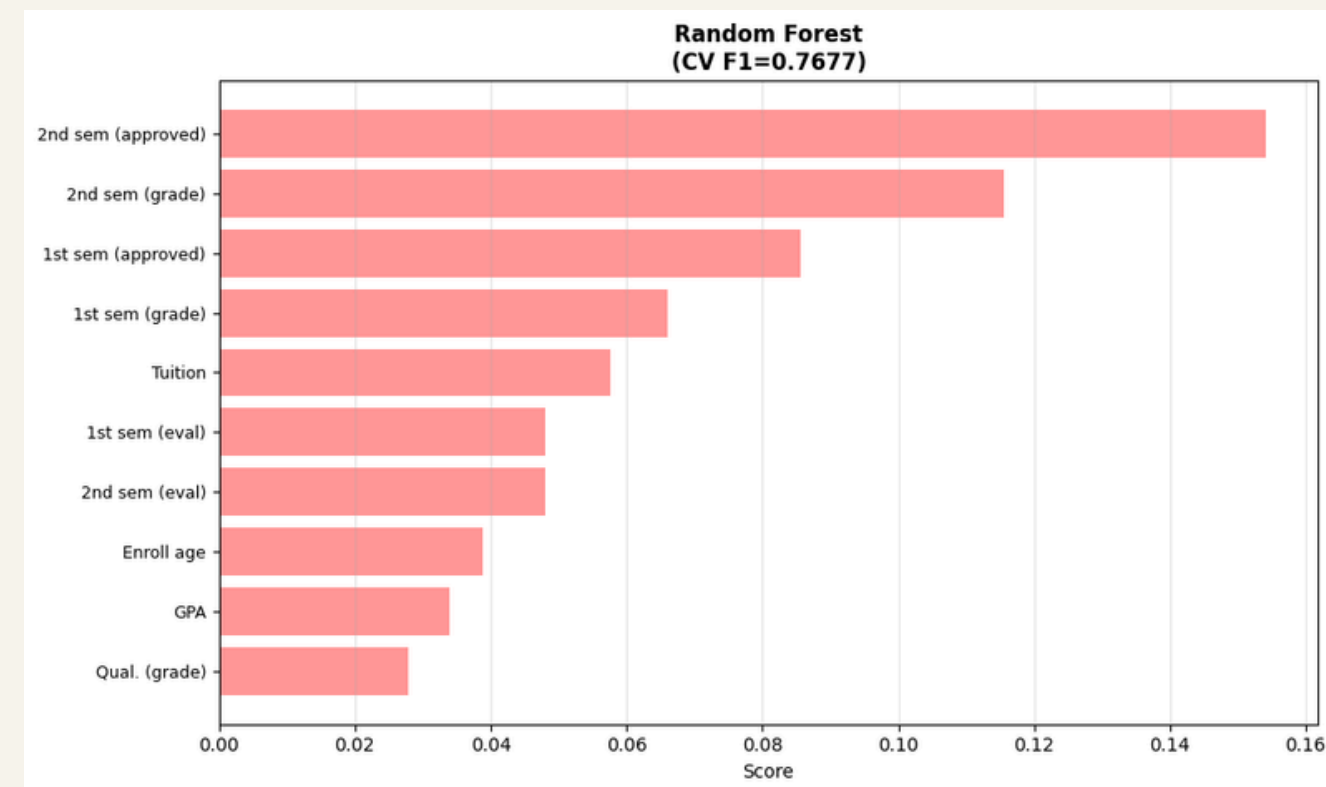
Dataset	Test Accuracy	Test F1	Test ROC-AUC
Early	0.617	0.587	0.738
Mid	0.734	0.714	0.857
Full	0.762	0.752	0.882

Results

Feature Importance



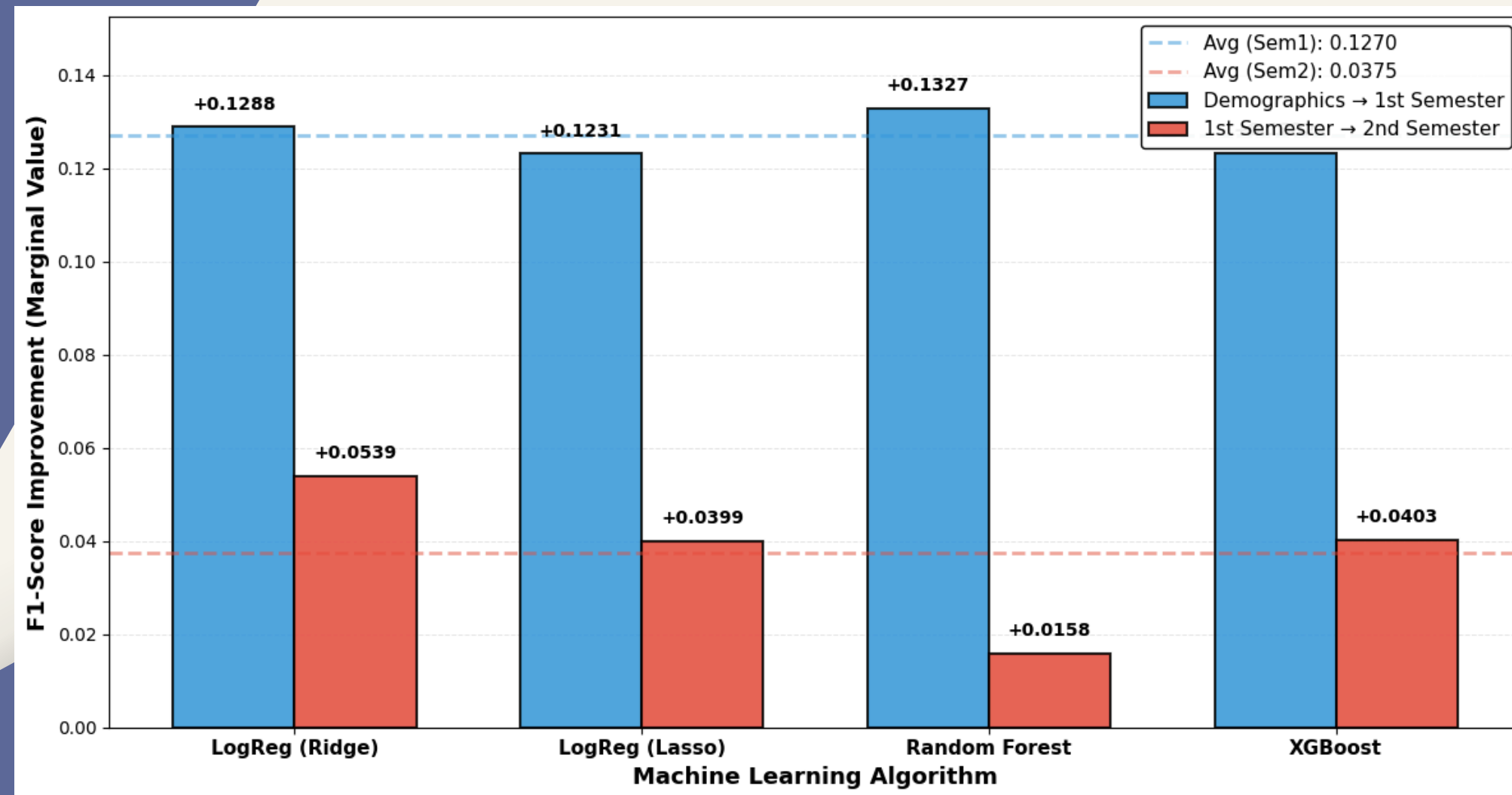
Feature Importance



1. Academic metrics are the strongest predictors (60%+ importance), both in 1st and 2nd semesters
 - Units approved, grades: 35%
 - Other academic (eval, w/o-eval): 25%
2. Demographics and financial status: 35% combined

Results

Incremental Contribution: 1st vs. 2nd semester

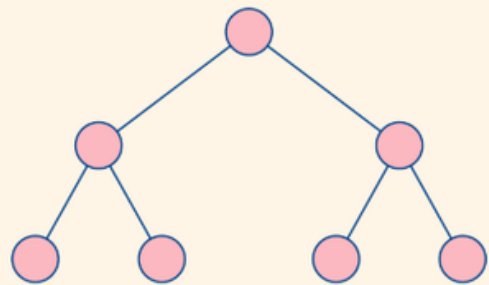


- Both RF and XGBoost show consistent benefits from the progressive datasets
- Average accuracy for demographic data alone is insufficient (F1 61.67%)
- Adding 1st semester leads to an 11.7% increase (collect 1st semester grades for early intervention).
- Adding 2nd semester only adds 2.8% marginal value (diminishing returns).

Discussion

There were three major takeaways from our project

XGBoost Algorithm



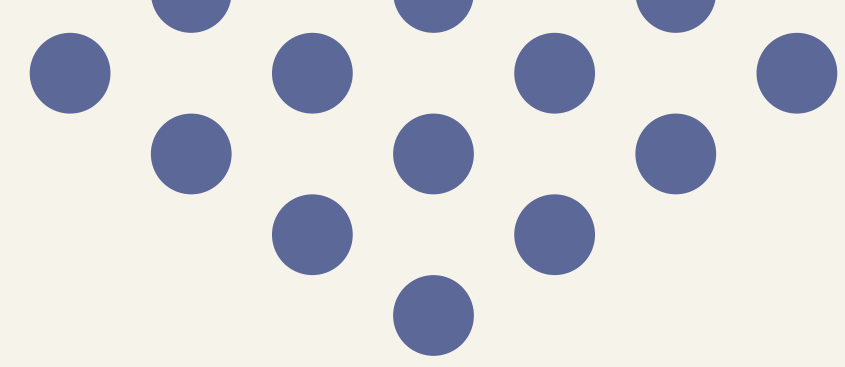
XGBoost models performed the strongest (77.51% accuracy, 0.7672 F1, 0.8943 ROC-AUC), suggesting a strong non-linear relationship between features and that regularization improved overall model performance



For the XGBoost model, the most important features were directly related to academic data (e.g., credits completed). However, tuition was also an important feature, suggesting that socioeconomic factors may also impact dropout risk.

SEMESTER 1

We recommend deployment of XGBoost model with only first semester data, as the marginal improvements of adding the second semester data are low (and time-consuming).



Conclusion

- The model performance demonstrates how ML can be effectively applied to predict student drop-out rates, with tree-based methods (e.g., XGBoost) being the most effective strategies overall.
- The data utilized for this project only includes students from Portugal. Further research is needed to predict dropout rates for universities in other countries with different educational systems, subject to cohort change, policy change, and economic environment.
- For universities looking to implement their own analyses, we recommend monthly accuracy checks, frequent retraining (when new data become available), and quarterly feature importance reviews.



References

- "Education pays, 2024," Career Outlook, U.S. Bureau of Labor Statistics, May 2025
 - Lochner, Lance, and Enrico Moretti. 2004. "The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports." *American Economic Review* 94 (1): 155-189.
 - Lochner, Lance. 2020. "Chapter 9 Education and Crime" *The Economics of Education* (Second Edition), 109-117.
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QUESTIONS?

